

Phonetic Drift vs. Phonetic Attrition

– same phenomena, different names?

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Roadmap

1. Personal background
2. Drift vs. Attrition
3. Investigating Drift and Attrition
4. Learning mechanisms

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Personal background

I'm a doctoral student in General linguistics who investigates **learning mechanisms** in phonetics, phonology and morphophonology



Supervised by Ruben van de Vijver (Heinrich Heine University Düsseldorf) & Volya Kapatsinski (University of Oregon)

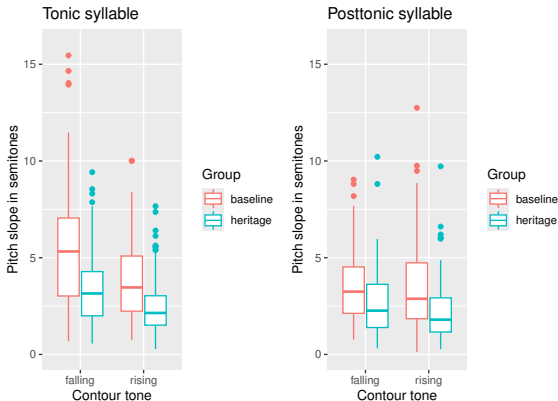
- So why am I giving a talk on phonetic drift and phonetic attrition?
 - Interest and past research
 - Learning theoretical perspective (future directions?)

Personal background

- First confrontation with phonetic drift/attrition during my bachelor's thesis (2019)
 - Thesis: Reciprocal transfer effects of suprasegmental features in Serbo-Croatian heritage speakers (Mujezinović, 2019)

Personal background

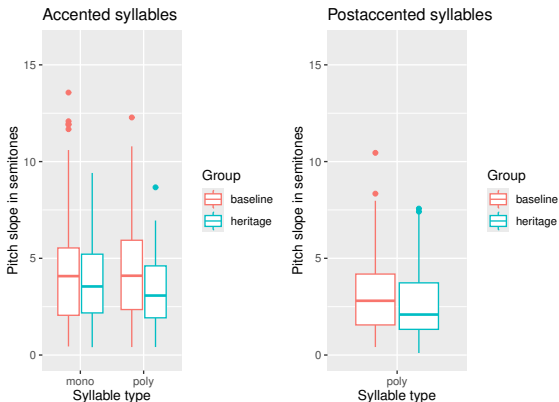
Realization of Serbo-Croatian pitch accent by L1 Serbo-Croatian, L2 German speakers



Less extreme pitch movement in heritage speakers (as shown by pitch slope)

Personal background

In comparison: Realization of German word accent by L1 Serbo-Croatian, L2 German speakers



- Comparable pitch slopes between baseline and heritage
- difference between mono and poly in heritage speakers only (not important for this talk)

Personal background

What is the nature of these effects?

- Is this drift? Is this attrition?
 - Was there drift first?
 - How much is related to priming? i.e. heritage speakers were recruited in a German setting
- Is this phonetic? Is this phonological?

Goal of this talk: Discussing commonalities and differences between phonetic drift and phonetic attrition

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Definitions

Phonetic Drift

[...] phonetic changes that occur in one's native language (L1) due to recent experience in another language (L2), a phenomenon referred to as PHONETIC DRIFT (Chang, 2019).

- Changes that affect phonetic categories of the first language
- What phonetic drift is **not**:
 - Changes in register (e.g formal vs. informal)
 - Speech accommodation to interlocutor
 - Phonetic Attrition?
 - Priming effect?

Definitions

Precondition for drift

These results thus suggest, in line with the SLM and Chang 2013, that drift is **dependent on the disparity** between L1 and L2 sounds falling within a **'sweet spot' of cross-linguistic distance** (i.e., not so small that there is no room to drift, but not so large that the L2 sound is no longer in the phonetic neighbourhood of the L1 sound) (Chang, 2019).

Definitions

Phonetic Attrition

[...] studies into phonetic attrition are considered to reveal long-term changes in the native language pronunciation of an individual as a result of the acquisition of a new dialect or language acquired post adolescence (de Leeuw, 2019b).

Difference Drift vs. Attrition

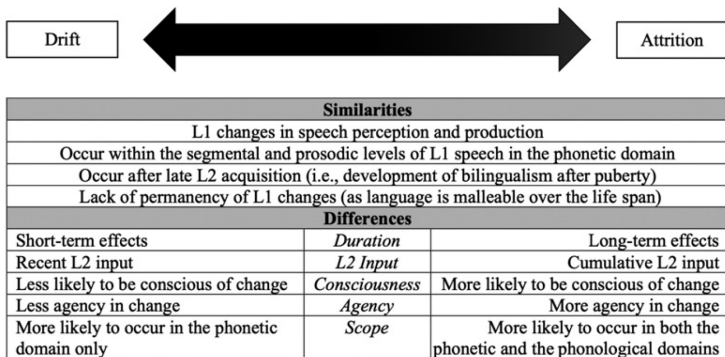


Figure 32.1 Theoretical similarities and differences between drift and attrition.

Differences Drift vs. Attrition (de Leeuw & Chang, 2024)

Difference Drift vs. Attrition

Continuum

Emerging longitudinal research may reveal how such long-term phonetic changes in the form of attrition are affected, or perhaps governed, by such short-term changes, or **whether there is a cut-off between the two at all** (de Leeuw, 2019b).

Potential factors

- L1 specific factors
 - Quality of L1 use
 - Codemixing and Codeswitching
 - Education in L1
 - Amount of L1 contact
- Sociolinguistic variables
- Non-linguistic factors (domain general)
 - e.g. Inhibitory control

Reversibility

Changes in voice onset time (VOT) in oral stops before changes in vowel formants (Chang, 2012; Lang & Davidson, 2019)

- VOT and formants can revert after L1 reimmersion
- Certain properties revert faster once L2 use decreases
 - VOT reverts faster than formants or pitch
 - Prosodic changes last longer

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Investigating Drift and Attrition

- According to de Leeuw (2019a): longitudinal studies on L1 changes in adults are rare
- Studies have to account for several factors

Investigating Drift and Attrition

Some effects that need to be taken care of (based on discussion in de Leeuw (2019a)):

- Frequency effects: How frequent is the contact to home country?
- Other explanations: Did the language of the community change? And were these changes adapted?
- Formality: Do interlocutors know each other well? (in interview scenarios)
- Socio-indexicality: Are some sounds less prone to changes due to marking social identity?
 - Some markers are kept by second generation heritage speakers

Investigating Drift and Attrition

- In cohort studies: Late bilinguals show strong or weak effects of L2 on L1 **at test time**
- Unclear, if effect is due to age of acquisition or length of stay (cumulative effect)
- Short term and long term effects are testable

Investigating Drift and Attrition

In what circumstances do drift/attrition have a particularly strong effect?

- First years of immigration
- Weaker effects if speaker has previous experience in L2
- But changes take place even in those cases (e.g. de Leeuw (2019a))

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Learning mechanisms

Usage-based linguistics

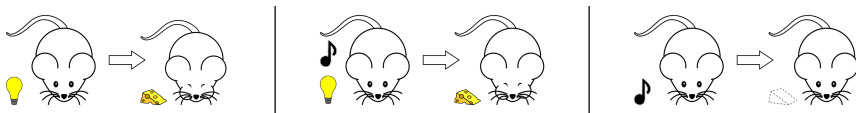
Usage-Based Theory looks at the way experience with language directly impacts the cognitive representation of language. Under its purview is usage of all types, including everyday social encounters, acquisition and **learning**, and changes in language use over time and space (Bybee, 2023)

Change in focus: Investigating learning mechanisms in drift/attrition

Learning mechanisms

Discriminative (error-driven) learning (Harmon, Idemaru, & Kapatsinski, 2019; Mujezinović, Kapatsinski, & van de Vijver, 2024; Nixon, 2020; Nixon & Tomaschek, 2021; Olejarczyk, Kapatsinski, & Baayen, 2018; Ramscar, Dye, & McCauley, 2013; Ramscar et al., 2010; Rescorla & Wagner, 1972)

- Prediction and prediction error: How well a cue predicts an outcome matters
- Positive evidence: Associations increase if a **cue is present** while an **outcome is present**
- Negative evidence: Associations decrease if a **cue is present** while an **outcome is absent**



https://en.wikipedia.org/wiki/Blocking_effect#/media/File:Classical_conditioning_-_blocking.svg

Alternative

ADAPPT theory: L2 contact and acquisition **always influence** L1
(de Leeuw & Chang, 2024)

- Consequence of bilingualism
- Other languages in a speakers inventory have to adapt to a new variety to allow to easier incorporate the new variety

Alternative

Radical alternative view: There is no difference between drift and attrition

- Both amount to difference in cue-outcome weighting
- Serbo-Croatian pitch data: more use of pitch excursions when immersed in native language context
- Pitch movement is less discriminative when immersed in a German setting
- Learning effects account for edge cases and clear cases
- Compatible with ADAPPT

Discussion

We now have a better idea about how drift and attrition differ – **but** there is enough room for discussion:

- How can we ensure that we are **really** dealing with attrition when getting our data?
- How can we ensure that we are **not** dealing with priming/speech accommodation effects?
- How can we ensure that we are investigating both phenomena in a valid and reliable way?
- Is it possible to avoid the distinction and to subsume both phenomena under lifelong phonetic changes?

Own opinion

- Difference between phonetic drift and phonetic attrition more like a continuum than clear cut categorical differences (similar to de Leeuw & Chang (2024))
- Phonetic drift for general changes in L1 more appropriate term
 - Reserve phonetic attrition for more severe cases
- But: we should also avoid using phonetic drift for effects where priming or speech accommodation are sufficient
- Lifelong language change and malleable phonetic categories are **learning effects**

End

Thank you for your attention!

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Bonus

Two further examples

Assimilatory Drift in Production

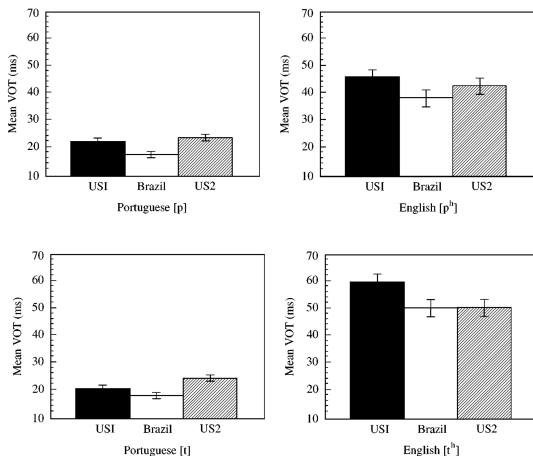


Figure 1: Phonetic drift in stops (Sancier & Fowler, 1997)

Effects of language and recent exposure (+ interaction for dental stops)

Dissimilatory Drift in Perception

- Kellogg & Chang (2023): Perception of voicing contrast in L1 may shift after less than one hour of input
- Perceptual shift independent of attention
 - participants subconsciously adapted even in distracted condition
- Dissimilatory effect
 - Voiceless oral stops in L1 have to show longer VOT to enhance contrast to oral stops in L2